Ensuring Privacy and Cyber Safety in the Online Gaming World for Children

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2022 International Conference on Cyber Resilience (ICCR) | 978-1-6654-6122-1/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICCR56254.2022.10024690

Abstract— **Gaming became one of the most popular activities during the pandemic. Most video game players are children since they have better access to devices connected to the internet. This has led them to spend more time playing online games. Games provided enjoyable ways for them to maintain social contact and helped them cope with the effects of lockdown. Children were not monitored while playing games and had access to strangers and downloaded many games, resulting in several data breaches and cyber-attacks. Children are less aware of the risks associated with online gaming, but it is a growing target for cybercriminals due to the amount of data and money involved. In our research, we provide promising directions for promoting children's awareness of these risks and their ability to mitigate them.**

Keywords— Online Gaming, Children Awareness, Cyber Security & Data Privacy, Human Factors

I. Introduction

The Covid-19 pandemic has impacted every walk of life. Lockdown restrictions affected the way how we live and interact, shop, study, work, socialize and so on. Day-to-day activities and interactions during these times have raised concerns about wellbeing. Digital platforms came to the rescue as strict lockdowns resulted in unusual circumstances. Citizens shifted to digital options, these range from video chat, online conferencing apps, health and fitness, religion, online ordering and delivering, media and entertainment, e-learning, and gaming. Digital access is grown exponentially. Global Statshot report was published by global in partner with we are social and Hootsuite [1] reveals there are 7.93 billion population on the planet as of Apr 2022 depicted in Fig 1 and out of which 5.32 billion users use the mobile device this is equal to 67 per cent of the entire population; the current statistics shows more than 63% population is accessing the internet and 58.7% of them have a social media presence. With widespread internet and mobile access. The challenge here with all this digital explosion is that many of them do not have the right awareness of how to use devices and how valuable their personal data is when it comes to the digital world and when it comes to children it is all the way more.

Fig 1.Global Digitial Presence Source: Global Weindex

A. The Need for Digitial Presence

There are several advantages of digitization, more than 5 billion people around the world have access to the internet. During the pandemic, many children were able to attend classes using their devices and many people from rural areas were able to access the internet and order goods and services for their daily needs. Due to strict covid restrictions, all of them were asked to be at home and this might have impacted everyone in greater ways of depression. These online platforms supported mankind as they were able to stay connected and communicate with their family and friends. At some point in time, people were even scared to step out of the home even to see the doctors. Online consultation is offered that supported many patients. Business and daily life were not impacted majorly as digital space had become the primary space. From Fig 1 it is very clear that Smartphone is one of the primary ways of accessing the internet and staying connected compared to other modes it is also affordable with a wide range of mobile phone companies offering smartphones at very reasonable and affordable prices. Comparative to other modes of accessing smartphone is the preferred way to access the internet and social media. India recorded about 365-million online gamers during 2018-to 2022 published by the Statista research department [1] this was estimated to reach 510 million by 2022.

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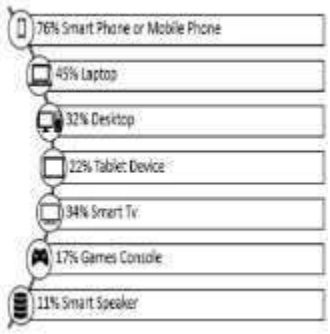

Fig 2. Time Spent on Devices Source: GlobalWeindex

B .Online gaming - A double edged sword

Online apps are termed a double edge sword as it offers many benefits and advantages, it also has severe disadvantages when they are misused. The apps are several categories starting from educational, quizzes, phonics & reading, spelling master, mind games, board games like chess math puzzles and riddles and gaming. All these apps should be used with care. When it comes to gaming apps, they have several features like rooms and allow multiple players to play together. Such games need to be monitored and to be played with care and consciousness to encourage the positive side of them.

II. Literature Survey

Chen Zhao 2018 [2] in his work he highlighted a few of the cyber security issues on trojans, man-made hacking server maintenance problems and he also suggested and protentional solutions on well - build encryption systems, high-performance server or bandwidth. In the digital era, the bandwidth and server performance are not challenging with the affordable high-speed internet and server spaces as per the usage methods. This work highlighted a limited number of issues and it also suggested the need to further work by extending the scope as there are many new trends in the field of cyber security and security issues are evolving daily.

Andhika Cahya Kharisma, Rizki Fitryasari1, Praba Diyan Rahmawati 2020 [3] in their study they have analyzed 111 respondents from Indonesia on the relationship between online games and addiction. Addiction is not healthy, and it leads to many other behavioural challenges. Duration of the games is a major concern here and it leads to addictions. Game addiction leads to motivational monopoly, and it is majorly impacting their sleep quality and sensory system, all these might impact their growth

Smahel, D., Machackova, H., Mascheroni, G., Dedkova, L., Staksrud, E., Ólafsson, K., Hasebrink, U 2020 [4] have done extensive work by collecting information from 19 European countries aged between 9 to 16 years. There is a substantial increase in the usage of smartphones in children and this had doubled in a few countries, and children above 15 – 16 access the internet and devices on daily basis. Girls tend to socialize more. Most of the children responded something online had upset them or bothered them, they had rarely shared these experiences with their teachers, parents or even their friends in most countries, around twice as many boys as girls play games online daily and girls responded that they spend too much money on games overall and this is also leading to misuse increase with their age. Smartphones are the preferred means of gaming online. online games are one of the top lists of activities done daily. Work also highlights the instances where there is major data misuse like creating pages about a person i.e impersonation and someone using passwords and they were able to find their location by using their mobile. Their devices are prone to spyware and malware due to the higher usage of online games.

Daniel L. Kingdaniel, and Paul H. Delfabbro 2020 [5] in their work highlighted pandemic lockdown mandates and quarantines that have increased online gaming activities and led to heavy consumption of digital entertainment. Online games promoted socializing, achieving positive outcomes in times like pandemic stress reduction. Excessive usage of gaming harms mental health and impacts sleep patterns [3] and physical health. At times it is difficult to identify due to the work and school practices. Considering the positive side it is important to promote balanced approaches when it comes to gaming to support physical wellbeing

Catherine Knibbs 2021[6] in her work she shares her extensive experience on the behaviour in cyberspace especially on children using devices, she extensively covered aspects of social media usage and abuse and the impact of unmonitored screen time and the extreme behaviours because of those.

Anand n Raut's 2021 [7] work highlights the greatest threat lies in applications that enabled to the collection of personal data of children posing several kinds of data privacy threats. He highlights the need for regulators to intervene and relook into the existing law framework.

III. Motivation

The above-mentioned research efforts have focused on addressing challenges and recommendations in cyberspace and partially on a few of the challenges of online games for children. As the technology keeps evolving it leaves behind a gap in terms of security and privacy. There is a constant increase in the number of cybercrimes reported and there are a greater number of unreported cases [7] where children are suffering in silence so there is a bigger need to come up with the best approaches to address these challenges. This work highlights the challenges in online games from security and privacy and human behavioural point of view and proposes ways to protect against them.

IV. Challenges in Online Gaming

Playing games is essential for children's healthy development. Additionally, they foster creativity, imagination, and cognitive and emotional strengths as online games are attracting more kids and offering all the above benefits while providing larger spaces for them to learn and grow. They are helping kids to

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socialize improve analytical skills, gain teamwork skills, multi-tasking, and better eye-hand coordination. Research also shows that playing frequent games will improve memory power and helps as a brain booster [6]. Imposed lockdowns resulted in no access to playgrounds and public places, this led to a rise in the use of devices for various purposes like entertainment, communication, gaming, education, e-commerce etc. There is exponential raise in children [4] accessing digital platforms and they also have become easy targets for cyber attackers.

The advantages of online games for kids are also multifold, but the challenges are also numerous.

Fig 3. Online gaming current challenges

A. Cyber Security

In recent years, cyber security issues in online games have been on the rise. Common cyber security issues in the game include information disclosure and cyber-attacks [8]. Video game revenues are on the rise as a number and in India alone [9] there is a growth rate of 165 % between 2016 to 2018. There was a survey carried out by IDC [15] on revenue generated on several types of video games and these numbers do not include ad revenues.

Table 1: Video Game Revenues in billions Source : Market Watch

Year	Mobile	Pc	Console
2018	8.70%	3%	13.60%
2019	15.40%	6.20%	-2.60%
2020	24%	11%	19.60%

As depicted in Fig 4 we have several cyber security challenges

a) Cyber Bullying -Cyber bullying unfortunately takes numerous forms, but some are specific to gaming platforms. Bruiting is employed by cyber bullies to shoot hurtful and dangerous dispatches on to players, or to spam global converse channels with obnoxious commentary about their victims

Fig 4. Challenges associated with Cyber Security

b) Phishing Scams -This is gaming version of phishing scam where user is tricked to give away their account details. Fraudsters send email asking them to substantiate their login credentials resulting in stealing gamers account information. After gaining access fraudster can sell entire account, spread phishing sites to other gamers, steal credit card information to withdraw funds.
c) Online Predators-They befriend with children on online gaming platforms and share game tips sometimes they also share game currency to advance in the game. Few games include team play in these they can communicate in chat rooms. Children are isolated from their parents and real-life friends by predators who take on the role of the "only person who really understands them.
d) Hidden Fees-Most online games use the freemium model, in which the basic content is free. To access advanced features, a fee is required. Several games offer subscriptions and other features such as avatars, virtual currencies, weapons, and special skills to advance in the game in exchange for payments
e) Malware -Malware can be inserted into the game by adding malicious code to the game, this code is used as back door to access personal and payment information, fake mobile versions of popular online games also install malwares on devices.
f) Software vulnerabilities -Gaming software is prone to security issues, vulnerabilities are exploited from time to time posing risks to players. Most of the vulnerabilities are patched and released. In such cases gamers have nothing much to do only to patch as and when required
g) Ransomwares -Gamers invest a great deal of time into their online characters. From time to time, gamers purchase powerups from third parties to benefit their characters. Third-party unlicensed powerups are often

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injected with ransom. Players pay substantial amounts to regain access.

h) Unprotected Game Servers -Hackers can breach any companies' servers and gaming servers are not prone to similar attacks. When gaming database gets traduced hackers earns access to several thousands of gamers, it is important to limit the information shared on these platforms

B. Data Privacy

There are several issues surrounding online gaming privacy. Online games have become the primary mode of playing and children are spending more time online than playing offline games. Mobile devices have become primary sources for children to go online [10]. They download apps from the play store and most of the apps seek permission to access one or the other dangerous permissions depicted in Fig 5. These permissions compromise personal data.

Fig 5. Dangerous permissions in online apps

Data is the new currency when it comes to hacker's world as well, so they access several types of personal data using these apps [11]. Survey was conducted by Arrka during 2021 on state of data privacy of Indian mobile apps [14] and few findings were depicted below

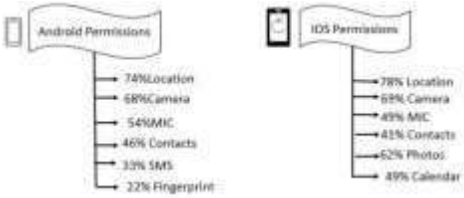

Fig 6 Top permissions on different mobile platforms Source: Arrka

Both IOS and Android operating systems poses high risks when it comes to critical permissions access. Indian gaming industry is witnessing growth at constant way on the other hand Indian children apps have one or the other dangerous permissions. Age-inappropriate content is displayed on some of the apps, and this is worrisome.

Fig 7 Indian Children App's Source Arrka Survey -2021

a) Webcam worries - Most of the digital devices come with inbuilt webcams and these are easy targets for hackers. They control webcams remotely. Multiplayer online games pose many risks as children use video chatting while playing games with strangers.

b) Location Tracking - Some online games access location information and sell this information with advertisers. It will be more dangerous if the same data is exploited resulting in security breach.

c) Microphone access & Voice Recognition - Few multiplayer games support conversations between players and children, at times may not realize the microphone is transmitting their entire conversation to strangers as they are totally involved in the game posing privacy threats. Voice recognition technology can be cloned evading security controls with cloned voice samples

d) Personal Information on Console and devices - Computer game console stores personal data, these data must be wiped before recycling or selling these devices. This will pose many dangers if they are are not erased and falls into hands of strangers

e) Keyloggers - These are kind of malwares that are used to sniff keystrokes to capture all login credentials, they are shared via emails or message chats pretending to offer free access or offering some points in the games. Hackers gain access to account information using this.

f) Invading self-expression - Online games offer several avatars and children feel so attached to these avatars as they match their personalities. Avatars reveals players choices, most games share context of relations with social network sites such as third party, in such instances children feel invaded of their self-expression as they can be behaviourally tracked easily

C. Behavioural issues in children, issues relating to human behaviour

There is close relation between mobile device and gaming addiction and mental health issues in children, it is also noticed that online game addiction leading to depression, social anxiety, and loneliness [10], many research and studies have been performed on this subject. Gaming addiction is the root cause of behavioral issues in children as depicted in fig 8.

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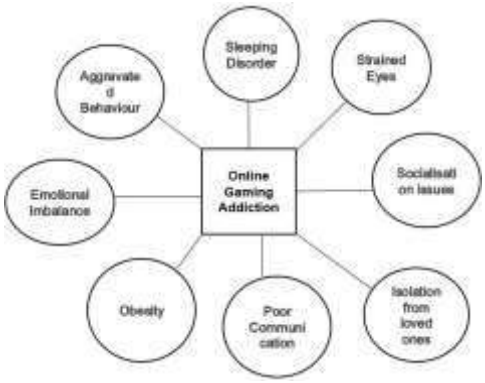

Fig 8 Issues relating to online gaming addiction

a) Gaming Addiction - This is one of the major issues associated with the use of digital devices and playing online games. Researchers have found that internet game addiction can lead to problematic behaviors, such as impulsive failures [01]. Playing violent games that involve killing and destroying will lead to aggressive behavior in children

b) Health problems- Online games are heavily impacting children's health, and this is one of the high risks. Many children are falling into obesity as they do not have any physical movement with long hours of playing games. This is also straining their eyes and many children must have glasses from childhood. All these conditions are due to excessive amounts of time spent on the internet. As depicted in Table 2 [1] younger generations are spending more time on internet when compared to adults.

Table 2 : Time spent on internet Source : Global Stat Short Report

Age Group	Gender	No of Hours Spent
16-24	Female	8.00
	Male	7.22
25-34	Female	7.16
	Male	7.09
35-44	Female	6.40
	Male	6.35

c) Emotional Imbalance-Talking to strangers and having so much glued to online world they do not show any interest to spend time with loved ones and from time to time they have emotional imbalances and exhibiting impulsive behaviors

II. Recommendations and Best PractiCes

The dream approach to protecting our younger generation is to sensitize them on a regular basis about digital threats. As the challenges are multifold, the recommendations and best practices must be followed at various levels. Besides, technical and regulatory support is essential for a safer gaming environment.

Fig 9 Best Practises assocaicted with Online Games

A. *Technical Recommendations*

i. Install antivirus and periodically scan for viruses and malwares and avoid using pirated versions of games
ii. Configure privacy settings, turn off location tracking and from security settings turn on two factor authentication
iii. Do not click or open links and attachments if you receive from strangers during the game
iv. Camera/webcam default settings must be set to off, if necessary, use physical shields /covers or even a piece of paper
v. Never add your card details to freemium games, it is always a good practice to activate purchase password from account settings menu. As a good practice regularly check credit card bills to ensure only approved purchases are being charged
vi. Perform extensive research about the game developers and vendors, and read recent reviews
vii. Never share personal information, use different names and passwords across different accounts and platforms
viii. Update device and games on devices regularly
ix. While installing any games on your mobile, check what permissions it asks for
x. Watch out for any suspicious activity on accounts, contact customer support immediately
xi. Take screen shot of any offensive conversation or comments and report it to game admins
xii. Change passwords frequently
xiii. If you think your data has been compromised, ensure to verify the same and report it

B. *Regulatory Recommendations*

CERT In had published number of cybercrimes reported during last year's [14] table illustrates cybercrimes are on the rise year by year, posing a serious challenge. Technical measures are not the only way to combat cybercrime. Regulations also play a key role.

Table 3: Reported Cyber Security issues in India Source Cert In.

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Year	Cyber Attcaks
2018	208456
2019	394499
2020	1158208
2021	1402809
2022	212485

a. Online gaming platforms offering services to children and collecting personally identifiable information must not monetize data and they must delete data as soon as they leave platform.
b. Local laws should mandate privacy by design approach for platfroms collecting personal data. If such laws are not enforced the platfroms must come up with self regulatory practices and standards of security & privacy
c. There is no clear classification between gambling [12] games and normal games and the lack of this is also making it hard for elders/parents to forsee peontial
d. Parent consent has to be mandated by law via email or mobile phones

C. Human /Societal Recommendations

In addition to these technical and regulatory measures and recommendations parents' schools and society must play their part to sensitize children

a. When it comes to children playing online games parents have bigger role to play, they must ensure children discuss and they approve all their kid's game downloads.
b. From time-to-time parents must discuss with kids on online gaming risks like bullying, identity theft, personal and financial data theft and several ongoing scams and frauds
c. Along with academic education schools have to offer digital space awareness to children as this space needs not much introduction only, they must understand the risks associated and must be bold to report in case of any issues bothering them
d. Best way is to open cyber clubs and have ongoing activities on weekly basis for children of all age groups
e. Establishing conversation at early ages will help them to be prepared and avoid threats.

III. Acknowledgements

We thank Arrka for generously letting us use their survey results and Macaire Odalis Reddy for useful discussions. We also thank our institutions The Foundation School and SAP Labs India Pvt Ltd for allowing us to work on research topic on online gaming that would help the wider audience in safeguarding children.

IV. Conclusion

The number of online gamers is growing every year, and this industry is not yet highly regulated. These platforms target children and women. Our efforts to provide a holistic approach will enable many children and educational institutions to raise awareness. Gaming companies also must incorporate security & Privacy by design principles as best practices. Further work on effective technical approaches to protect and respond in case of cyber-attacks will be greatly beneficial.

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